

Ideas Parked — Not Yet Merged

Staging document; each entry marks its eventual destination

This document holds ideas discussed against Pattern Arithmetic v15.19, Inference, and Assembly, none of which have been written into any of those three files yet. Each entry states its status and its intended destination. Nothing here is settled; the point of this file is to let that be true without leaving Paper 1 mid-edit.

1 Destined for Pattern Arithmetic (still formally open — but a leading resolution candidate has emerged, with a citation)

Mix across differing directions. The paragraph’s own text still reads “left open.” Nothing here changes that text. What changed is that the paper already contains a worked, concrete instance of this exact question, in a section not previously connected to it.

[Found: the paper already answers this question once, for one sort, and points toward a general answer.] The *chladni* inclusion states, word for word: “Across modes the map is not available, and not for want of effort. Mode space is not closed under addition... Cross-mode coexistence is therefore $\oplus$ — a bowl of modes — and the resulting figure is a reading of that bowl which is not $\boxplus$. This is the open question of [the differing-directions open item] met in a concrete domain, and it suggests the question may be closed negatively for this sort rather than left waiting.” This is a citation, not a new argument — the paper’s own words, in a section that was not previously read as bearing on Gap 1 directly.

[Why this converges with everything else worked through in this project, not just *chladni*.] Once Mix is read as “resolve several reports on one shared question into a verdict” (argued in v2/v3, carried forward below unchanged), refusing across-direction combination stops looking like a gap and starts looking like the correct behavior of an operator that was never asked that question in the first place. And every practical workaround that actually worked in this project was already living on the $\oplus$ side of precisely this line, independently: bowls holding separate weather channels, per-channel similarity scoring, the candlestick summary for reference-atom building, Assembly’s own camp-to-system Hill bijection. None of them ever needed Mix to cross a direction. All four arrived, separately, at “keep it in a bowl, read it there” — which the *chladni* section had already stated as a formal result.

[Status: a leading candidate, not yet a decision.] This supports Route 1 (refuse outright, from the three already on file) over Route 2 (rotate first) or Route 3 (two-stage operator) as the likely resolution, with a real citation behind it rather than preference. Writing this into Paper 1 would mean editing the “Open: Mix across differing directions” paragraph to state the refusal explicitly, citing the *chladni* section as precedent. Not done here, on request — parked, ready, pending a decision to merge.

[Carried forward unchanged: the reading argued in v2/v3.] Mix is not “multiple vectors becoming one vector.” It takes several reports already agreed to answer one shared question and returns a verdict about that question. The shared $\hat{n}$ is the precondition that makes “these answer the same thing” true, not a boundary condition on a more general operator. Supporting

evidence: same- direction opposite-phase cancellation to black (principled contradiction-handling, not a fabricated compromise); unmixing’s proven ill-posedness, with metamerism as independent real-world precedent; the paper’s own opening line, “two notes yield a chord, not a larger integer... the word ‘add’ names a different rule.”

[Carried forward unchanged: the “we mix all the time” challenge, resolved.] Sugar in tea is one dial, never needed differing directions. Red plus green additively sits “at the same place” in the paper’s own words — already inside Mix’s settled domain. Paint-layer flattening is confirmed, across every source checked, as destructive and irreversible, matching unmixing exactly; compositing unflattened layers is $\oplus$, not $\boxplus$, which is why “unmerge them later” was always possible. Representing a whole painting as one vector remains Gap 1 itself, not a counterexample to it.

[Carried forward unchanged: the two-part split, now read as one part likely answered.] Embedding validity — does geometric closeness on a shared sphere track real semantic agreement at all — was the harder, prior question, true for hue by deliberate construction, unstated for an arbitrary sphere of concepts. Given the chladni precedent above, the likely answer for sorts where this was never stated is that no valid embedding needs constructing, because cross-direction combination is refused before the embedding question would even matter.

[Carried forward unchanged: still confirmed not to help.] Magnitude self-bounding near opposition does not license the direction that comes with it. The Hill form and Assembly’s Hill bijection both fix magnitude only, verified twice, never direction.

2 A correction, recorded so it is not repeated

τ_{weather} **names two different problems, and only one of them is Gap 1.** Raised once as a proposed admissible sort for combining different weather concepts (rain, drought) across directions — that question is genuinely Gap 1, and the discussion above bears on it. Raised a second time, later, meaning something narrower: given several already closely-aligned vectors within one bounded region, how to arrive at a single representative atom, using the Hill form. That second question does not invoke $\boxplus$ across differing directions at all — it is Assembly’s already-verified system-promotion mechanism (16 camps to 13 systems under the Hill bijection), recognized in a new setting, not a Paper 1 admissibility question. The two were briefly conflated in this project; recorded here so the same conflation is not repeated. Assembly’s version needs nothing further; Gap 1’s version is addressed in Section 1 above.

3 Destined for Pattern Arithmetic (advances an existing open item)

Location as a sort — a third candidate. The existing paragraph names two candidates and adopts neither. A third was tested here: dedicate separate, disjoint dimensions to content, to time, and to location, rather than letting any of them share an axis.

[Verified.] Two shifts acting through disjoint content dimensions commute exactly (checked numerically, not approximately: time-then-location and location-then-time landed on the identical point to full floating-point precision). Two shifts forced to share a single content dimension do not commute, confirming the disjointness is what matters, not rotation itself.

[Verified.] Content stays recoverable after both shifts: projecting back onto the content dimensions alone reproduced the original direction closely, while the time-axis and location-axis each carried a clean, separately-readable amount of shift, with nothing blended between them.

[**Open, unchanged.**] The encoding itself — what specific rotation a given time value or location corresponds to — is not stated anywhere. The geometry now supports it cleanly; the mapping does not exist yet.

This is real, checkable content, not yet written as an addition to the existing paragraph.

4 Destined for Pattern Arithmetic (open item stays open, unchanged)

Reducing a path by merging close actors. The existing paragraph asks for three things: a stated distance, a merge rule, and a bound on how far a reduced path’s endpoint may drift from the unreduced one. Discussion here supplied a distance (see below) and partially addressed the merge rule, but only for magnitude, not for direction — and direction is the harder half this item is actually about.

[**Verified.**] Switching the strength calculation from the companion paper’s clip to Inference’s already-established Hill form, $r' = |z|/(K + |z|)$, changes nothing about the resulting direction $\chi' = \arg(z)$. Checked directly on two constructed sequences with identical content and opposite order: both gave the same angle under either formula. The Hill substitution fixes evidence-count blindness in strength; it does not touch the order-blindness this open item is actually asking about.

[**Open, unchanged.**] No merge rule for direction exists. No drift bound exists for either magnitude or direction. This item remains exactly as open as it was before this discussion, now with the magnitude half more precisely separated from the direction half.

5 Destined for Inference (new mechanism)

Candlestick summary: a richer merge rule for reference-atom building. Not a resolution of the Paper 1 item above — a different question. Paper 1’s open item is about *walking*: can a shortened path still be walked to roughly the same endpoint. This is about *reading*: how to summarize a group of atoms for comparison against a reference atom, which is Inference’s own established business (its reference-atom section), not *T*’s.

[**Verified.**] Open/close/high/low, computed the way a trading candle is, distinguishes exactly the case flat Mix could not. Two sequences, identical content, opposite order — naive Mix gave the identical angle (47.5) for both. The candle’s open and close swapped between them ($20 \rightarrow 160$ versus $160 \rightarrow 20$), correctly reading one as ending badly and the other as ending well. High and low stayed identical for both, correctly capturing that only the *range* was shared, not the *direction of travel*.

[**Settled, matches existing machinery.**] The scope rule — every atom belongs to exactly one time unit, no overlap between buckets — is not new; it is precisely the discrete-time-bucket metadata already defined on a walk’s own steps in the companion paper.

[**Open.**] This gives Inference a genuinely better reference-atom merge rule than flat Mix. It is not proposed as a fix for Paper 1’s walking-specific open item, which remains untouched above. Whether open/close/high/low itself needs a coherence check analogous to $\bar{R}_c$ has not been considered.

6 Destined for Inference (naming caution, not yet a mechanism)

“Inference Mandelbrot” is not Assembly’s Mandelbrot, and should not share its name. Assembly’s multi-zoom promotes an unordered thing (a settled camp) and is fully handled by the Hill bijection already verified there. Whatever would promote an *ordered* thing (a walked path, a movie, a time series) into one higher-level atom is a different, harder problem — it is the direction half of the path-reduction item above, restated from a different angle. **[Naming flag only.]** If this is ever built, it wants its own name; reusing “Mandelbrot” for it would be the same kind of collision already caught and corrected for “collapse,” “singularity,” and the quantum swap test.

7 Destined across three papers (new, split by piece)

Meta vectors: a second kind of vector alongside information vectors. Proposed: an information atom may carry two distinct kinds of vector. Information vectors are Axiom 7’s bundle v_i ’s, capped at $r_i \leq 1$, potentially several sorts τ across a whole record. Meta vectors describe the atom rather than being part of its content — three named so far: V_{derived} , link vectors (pointers between atoms, one- or two-directional), and a timestamp vector. Explicitly left open whether more meta-vector types are needed. Meta vectors are not capped at 1, and grouping several atoms’ meta vectors together on like terms is proposed as a distinct mechanism for building higher structures. This is one idea that lands in three different places once traced to where each piece actually belongs.

[Destined for Pattern Arithmetic.] The taxonomy itself — information vectors versus meta vectors as a formal, named distinction sitting on top of Axiom 7 — is structural, the same kind of addition Axiom 7 itself was. V_{derived} already lives there. A timestamp vector, if adopted, would be a third, genuinely different treatment of time from the two already on file above: distinct from the discrete-bucket metadata already defined on a walk’s own step, and distinct from the dedicated-axis encoding in the location-as-a-sort entry. Not yet clear whether more than one of the three should coexist, or which.

[Destined for Assembly.] Link vectors specifically. This is the mechanism the shelf’s own neighbour-graph note was already naming without a concrete form: “a data structure on seats after rest, not T .” Grouping link vectors on like terms is collecting edges into a graph, not merging them — gathering, in the $\oplus$ sense — so it carries none of the open questions below it.

[Destined for Inference.] The grouping mechanism itself: querying several atoms’ meta vectors, confirming they sit close together, and promoting one shared meta-vector up to the group as a whole. This is the space-time gating idea from earlier discussion, generalized into a standing principle. Timestamp and location get *used* here; they are only *defined* in Pattern Arithmetic above.

[Verified, carried forward as a caveat rather than a new finding.] Combining several atoms’ V_{derived} ’s stays safe only under the constraint the original already carries: auxiliary, never promoted to a trusted, walkable atom on its own. Checked directly that neither Inference’s Hill form nor Assembly’s Hill bijection touches this constraint — both fix magnitude only, never direction — so grouping V_{derived} ’s inherits that same limit rather than escaping it.

[Note added here.] Link vectors were separately discussed as one possible mechanism for a limited, honest form of “uncollapse”: not inverting Mix (still impossible, unaffected by anything here), but keeping an explicit, external pointer from a collapsed atom back to the bowl that produced it, created at collapse time, before the bowl itself might otherwise be discarded. This is navigation via a kept record, not reconstruction from the collapsed result — the distinction the white-light/prism discussion turned on, confirmed there to matter.

8 Considered and found to need nothing filed

For completeness: the reference-atom and precursor-witness mechanism, the bundle-vector / uncapped V_{derived} distinction from Mix, the $\pi = (x; w_1, \dots, w_n)$ poem trajectory, and the precise definition of χ as internal phase-color (already stated in Axiom 1: “two identities may occupy the same direction without overwriting one another”) were all discussed at length and all turned out to already be fully present in Inference and Paper 1 respectively. Nothing from those threads is parked here, since there was nothing left unresolved to park.